

A Multimodal Pipeline for Personalized Recommendations from Greek YouTube Content: Integrating Automatic Speech Recognition and Aspect-Based Sentiment Analysis

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Abstract— Audio/video social media and video sharing sites like YouTube allow users to create their own community where they post audio content. This type of content includes product reviews which consumers use to assist others in making buying decisions, and this content can be processed using NLP and automated Aspect-Based Sentiment Analysis (ABSA) to produce input for recommender systems. However, developing these systems for low resource languages (such as Modern Greek) lags behind significantly. To address this gap, in this paper we introduce a complete ABSA framework, which transforms Greek multimedia reviews into structured and accurate data, to be used for personalized product recommendation. The proposed framework uses a multistage pipeline to address the complexities of spoken text, such as the highly inflected morphology of the Greek language and slang utilized in reviewing. The first stage of the pipeline transforms raw audio data into a form suitable for processing, while the second stage applies linguistic processing. Two alternatives were evaluated for this stage, where the first utilizes a native Greek Sentiment Lexicon and the second firstly applies Greek-to-English automated translation, followed by English Language Sentiment Analysis. The efficacy of the proposed framework has been assessed through an experiment in which each of the pipeline outcomes is compared against independent human assessments. The assessment shows that the translation-based approach achieves high accuracy and alignment with expert human raters. The proposed work also takes into account the effect of “noise” owing to Automatic Speech Recognition (ASR) errors and studies the propagation of these errors across 12 distinct product attributes. The proposed pipeline thus constitutes an effective solution for exploiting audiovisual content in recommender systems to generate personalized recommendations.

Keywords—*Aspect-Based Sentiment Analysis (ABSA), Personalized Recommender Systems, Modern Greek NLP, Automatic Speech Recognition (ASR), Multimedia Data Mining, User Preference Modeling*

I. INTRODUCTION

The digital revolution in how consumers access and process product information has propelled video-sharing sites like YouTube to the top of the list of sources of product-related information [1], [2]. Within Greece, “vloggers” and other technology enthusiasts produce highly influential review videos that provide the first point of contact for potential smartphone purchasers [3]. Additionally, these multimedia reviews contain unstructured data in the form of detailed opinions about products, which are much more descriptive than simply providing a numerical rating (i.e., stars) [4], [5]. Therefore, although these reviews are valuable within today’s digital landscape, there needs to be a way to transform them into structured insights that can feed into the algorithms used by personalized recommender systems (RecSys) [6], [7], [8].

Currently, many RecSys used by retailers throughout Greece do little more than recommend products to users who have purchased similar items in the past, using only basic metadata such as categories or ratings to make recommendations [9]. In cases where a consumer sets a specific set of requirements when searching for a product, the aforementioned recommendation method is adequately successful the majority of the time. However, this “black box” approach is unable to take into consideration the potentially differing weights that each individual user applies to various product features [10]. For example, for a professional photographer who also uses his mobile phone to take pictures, features related to the phone’s camera (e.g., sensor, resolution, zoom) are of major importance, while the phone’s design and cost are of less importance. On the other hand, for a college student, most probably the cost and battery life are of major importance, while features related to the camera are of lesser importance.

To achieve true personalization, a RecSys needs aspect-based ratings, which can be extracted from qualitative narratives found in expert vlogs [11], [12]. By converting these audiovisual streams into a structured Aspect-Sentiment

Matrix, a recommendation engine can effectively match a product's perceived performance with a user's unique, feature-specific priorities [13].

Performing sentiment analysis in Modern Greek to extract information for use in RecSys faces a number of challenges. The Modern Greek language exhibits high morphological complexity, manifested (among others) by the extensive use of inflectional endings and the presence of function words [14]. In this context, a lemma may appear in many forms, each corresponding to a different gender, number, case, or even a different part of speech (e.g., adverb vs. adjective). Since adjectives are the predominant means for expressing sentiment, their correct identification and lemmatization is of particular importance in the context of a pipeline that involves mapping terms against sentiment lexicons [15]. Another source of inaccuracy is the use of "jargon" (or "technical slang", such as "throttling", "lagging" etc.), which leads to increased rates of misinterpretation by machine translation tools [16]. This issue may be alleviated by pre-processing the text using machine translation. This work comparatively assesses this approach to determine whether the sentiment tone and strength in the original sentences are retained in the result of the translation and in the outcome of the pipeline.

The effectiveness of the process that extracts aspect-based sentiments from multimedia content is considerably affected by the accuracy of the Automatic Speech Recognition (ASR) stage. More specifically, this stage may introduce errors due to substitutions (wrong word), deletions (missed word), and insertions (added word) of words [16], and such errors may in turn lead to inversions of sentiment (e.g. the negation "is not good" being misrecognized as "is good"), or miscalculations of sentiment intensity (e.g. "is good" being misrecognized as "is so good"). In this paper, we elaborate on this issue by quantifying the amount of "noise" introduced during ASR and measuring the effect that its propagation [17] through the processing pipeline has on the final outcome, in terms of the Mean Absolute Error metric [18]. In this context, we determine the minimum quality requirements for the ASR phase, in order to maintain a reliable ABSA process, whose outcome may be effectively used for personalized recommendations.

In this paper, we address the issues listed above through the introduction of a pipeline that processes audiovisual review video input sourced from Greek YouTube, to produce structured, aspect-based sentiment recognition information. The efficacy of the proposed pipeline is validated through an experimental process comparing the outcome of the pipeline against ratings provided by expert human annotators. The main contributions of this paper are as follows:

1. We propose a pipeline for processing audiovisual review content in the Greek language to produce structured, aspect-based sentiment recognition data. This pipeline addresses the specificities and idiosyncrasies of the Greek language.
2. We comparatively evaluate two alternative methods for performing linguistic processing and sentiment recognition: one using a native Greek sentiment lexicon and a second employing machine translation and an English sentiment lexicon.
3. We evaluate the impact of noise injected during the ASR phase on the accuracy of sentiment identification.

The rest of this paper is organized as follows: Section II provides background information and reviews previous works related to this topic. Section III explains the design of our proposed system architecture. Section IV provides our experimental findings, and, finally, Section V summarizes the Conclusions, Limitations and Future Work.

II. RELATED WORK

Opinion mining has advanced from single-document-level polarity determination to much more sophisticated and feature-based systems [17]. Section II provides an overview of the basic concepts behind Aspect-Based Sentiment Analysis (ABSA), the specifics of the Greek linguistic environment, and the problems associated with incorporating ASR into recommendation processes [18].

A. Foundations of Aspect-Based Sentiment Analysis

Liu [19] defines Sentiment Analysis at the aspect level (commonly known as ABSA) as the process of determining the specific features of an opinion and the sentiment expressed toward those features. When reviewing products, general, document-wide assessments of sentiment frequently fail to identify conflicting sentiments within a review. For example, a reviewer may praise a smartphone's "Screen" while also criticizing its "Price." Many researchers believe there is an increasing trend toward developing greater granularity in data in order to gain better insight into consumer behavior [20]. Although recent successes using deep learning models have led to significant growth in usage, traditional lexicon-based methods continue to provide value for low-resource languages because they are transparent and require less computational power than deep learning methods when applied in real-time multimedia processing environments [21].

B. Sentiment Analysis in the Greek Linguistic Context

Effective sentiment analysis is highly dependent on the available NLP resources for the target language. The development of such resources for the Modern Greek language lags behind the progress made in other languages, predominantly English, owing to the limited developer community, as well as the language's unique characteristics including syntactic complexity and increased morphological variation [23]. The first sentiment analysis attempts for the Greek language used a combination of machine translation and English sentiment analyzers; however, recently effective native Greek sentiment analyzers have emerged.

GrAFS (The Greek Affect and Sentiment Lexicon) represents one of the most important contributions to Greek NLP [15]. It contains more than 32,000 words, each manually annotated with sentiment. These lexicons can now be easily incorporated into current applications thanks to modular tools, including spaCy [22] and the work in [23] on lemmatization for Greek. Even though great strides have been made in providing domain-specific resources for Modern Greek, there continues to be a lack of standard benchmarks for native versus translation-based models for Greek [24]. As such, many researchers utilize widely available datasets and tools, including VADER [11], which is a well-known tool for analyzing sentiment across large volumes of text.

C. Multimedia Mining and the Impact of ASR Fidelity

Multimodal mining (i.e., combining different types of input modalities, such as speech and visual) is an emerging area of interest in sentiment analysis research. Errors in ASR output (including substitution errors, where words are

incorrectly replaced, and deletion errors, where words are removed) can cause dramatic changes in the sentiment of the resulting transcribed sentences, including reversing the overall emotional tone of a sentence [25], [26], [27].

In Greece, the work in [24] provided early foundations for social media sentiment mining. However, the issues related to assessing sentiment in multimodal data remain largely unexplored. Therefore, our research investigates this issue by measuring how “sentiment decay”, caused by noisy transcription, affects subsequent steps in the NLP pipeline and ultimately impacts the accuracy of recommendation data [28], [29].

D. Sentiment-Driven Personalized Recommendation

Personalized RecSys represent a fundamental shift in the way that users interact with websites. To support personalized RecSys, feature-aware recommendation engines are being developed. Content-based filtering depends upon the attributes extracted from the content, and hybrid models that combine both quantitative measures (such as popularity) with qualitative measures derived from user feedback are becoming increasingly popular among contemporary RecSys designers [30], [31], [32].

For a recommendation engine to create truly personalized recommendations for users, it must understand what aspects of a product are of greatest importance to each individual user (for example, some users might prioritize battery life over design). Addressing this need, current research focuses on extracting detailed preference profiles from unstructured data, such as YouTube video reviews, to enhance the precision of personalized recommendations within the specific context of the Greek online market [33], [34], [35].

III. SYSTEM DESIGN AND METHODOLOGY

This Section outlines the core aim of the proposed system. The objective of the proposed system is to take raw, unstructured Greek audio/visual data from YouTube and create a structured Aspect-Sentiment matrix that can be used for recommending products or providing personalized recommendations. The process consists of a series of modules, which are outlined below. The overall structure is presented in Fig. 1, and it illustrates the modular approach to creating an Aspect-Sentiment matrix. These include:

- Data Acquisition
- Speech-to-Text Conversion (ASR)
- Linguistic Preprocessing
- Sentiment Analysis

A. Dataset Acquisition and Expert Annotation

The dataset collected for this research included twenty-three vlogs about smartphones. The data set has been populated using the top-ranked videos accessible via the Greek YouTube platform. Videos were selected for inclusion on the basis of (i) good quality audio, (ii) high levels of technical content, and (iii) potential to deliver aspect-based 'RecSys' input.

For the purposes of establishing a reliable ground truth, the data set has undergone a manual annotation process utilizing human expertise. A total of twelve key product attributes were identified and utilized in this manual annotation process. These attributes are: Camera, Battery, Screen, Performance, Design, Price, Software, Build Quality, Audio, Storage, Connectivity, and Overall Value. For each attribute, an expert expressed their opinion, based solely on the video content, as a number on a subjective scale; these opinions were then normalized to fall within the range $[-1, 1]$. The manual baseline is needed to (a) quantify the accuracy of the aspect-based sentiment detection performed by the proposed system and (b) assess the level of Sentiment Decay experienced due to the utilization of automated processes.

To improve the reliability of the manual annotation process, a double-blind annotation process was applied: each of the two experts formally reviewed each sentence, providing a score regarding the sentiment conveyed for one or more of the twelve designated attributes. When the two experts disagreed regarding the annotation of certain text segments with borderline sentiments or differing degrees of sentiment expression, they engaged in a systematic reconciliation phase, wherein they jointly examined the disputed text segment(s) until mutually acceptable agreements were established. The same procedure was followed regarding the overall scores assigned to the twelve designated attributes, based on the entire video text. The inter-annotator agreement was very high among all twenty-three smartphone reviews and across all twelve aspect (the normalized mean absolute error was .085); therefore, the mean annotator scores provided the final Ground Truth for evaluating performance.

B. The ASR Layer: Vosk Implementation

The first technical issue is transforming spoken Greek into processed text. The Vosk ASR engine, an offline Kaldi-based toolkit was used [36], [37]. Vosk was chosen over cloud-based systems because it supports the Greek language well and can be used to identify technical terms (i.e., megapixels, refresh rate) that are commonly found in a local area. An important component of our approach is the Dual-Track Error Analysis. Two versions of the transcript were maintained:

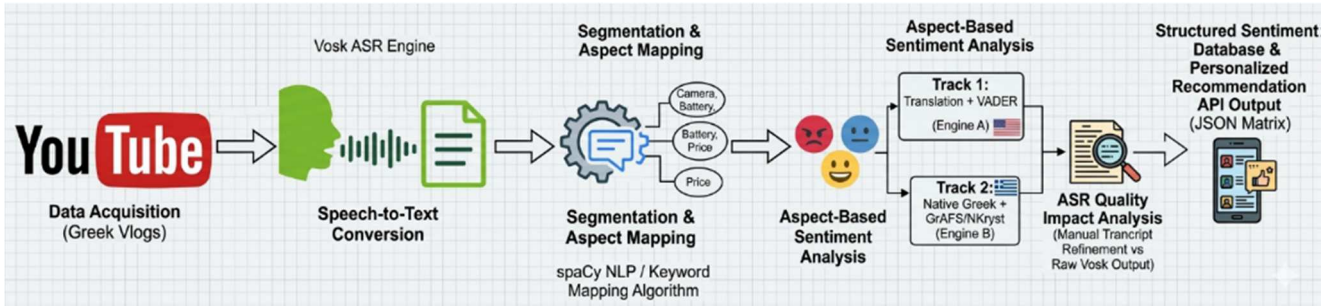


Fig. 1. The video aspect-oriented sentiment analysis pipeline with two tracks for sentiment quantification

1. Raw ASR Output: Unedited transcript output from the Vosk Engine.
2. Manually Refined Text: Cleaned-up version of the above; used to separate out the effect of the NLP engine's processing versus errors caused during transcription.

We evaluated both tracks to determine the types and magnitudes of error propagation that occurred along the pipeline that converts audible data to aspect-specific sentiment values. Our experiments demonstrated that under the strict acoustic filtering applied, the ASR model [37] exhibits errors that are owing to the very nature of tech-focused Greek YouTube, and more specifically to (a) domain-specific jargon, (b) high inflectional morphology and (c) English brand-name substitutions. Although the Word Error Rate (WER) metric is typically used to assess an acoustic models accuracy, in this work we focus on the impact of transcription errors on the semantics of the output text, which in turn drives aspect-based sentiment recognition. Therefore, rather than examining deviations at the lexical level, our evaluation focuses on the 'decay of sentiment', quantifying how insertions, deletions, or substitutions at the ASR stage affect the aspect-sentiment matrix at the end. This is a practical metric for the tolerance level of the downstream recommendation engines.

C. Linguistic Pre-processing and Morphological Analysis

Because of its high inflectional complexity, there are significant requirements placed upon modern NLP in order to successfully process Modern Greek.

The spaCy library [22] with the `el_core_news_lg` model (<https://spacy.io/models/el>) was employed to implement a customized pre-processing pipeline:

- Tokenization & Lemmatization: Each word is converted into its lemma (e.g., the adjectives “καλός” (good / masculine / singular), “καλή” (good / feminine / singular), “καλές” (good / feminine / plural) all map to the lemma “good”). This is vital for accurate lexicon matching.
- POS Tagging: The system identifies whether a particular word is a noun (possible aspect) or an adjective (sentiment carrier).
- Aspect Mapping Algorithm: A domain-specific dictionary was constructed where technical words related to products are grouped together under one of twelve broad categories. Examples of this grouping include technical terms, such as “Lens”, “Sensor” and “Aperture”, which are mapped directly to the camera aspect.

D. Dual-Track Sentiment Inference

The system compares how well it can extract sentiment using two different methods:

Method 1 - Translation-Augmented VADER (track 1): Greek lemmas are translated using Google Translate and then evaluated by the VADER engine. VADER is optimized for informal communication found on social media. VADER uses a pre-verified dictionary. Each item in the VADER dictionary has an average sentiment value as well as a standard deviation, providing the ability to identify sentiment at a very detailed level, as shown in Fig. 2.

Method 2 - Native lexicon-based inference (track 2): Utilizes the NKryst R wrapper to evaluate the Greek Affect & Sentiment Lexicon (GrAFS), eliminating the potential semantic shift that can occur when translating Greek terms into another language.

TOKEN	MEAN-SENTIMENT-RATING	STANDARD DEVIATION	RAW-HUMAN-SENTIMENT-RATING
abandon	-19	53852	[-1,-2,-2,-2,-2,-3,-2,-2,-1,-2]
abandoned	-20	109545	[-1,-1,-3,-2,-1,-4,-1,-3,-3,-1]
abandoner	-19	83066	[-1,-1,-3,-2,-1,-3,-1,-2,-3,-2]
abandoners	-19	83066	[-2,-3,-2,-3,-2,-1,-2,-2,-0,-2]
abandoning	-16	8	[-3,-2,-3,-2,-1,-1,-1,-1,-1,-1]
abandonment	-24	10198	[-4,-2,-1,-4,-2,-1,-2,-3,-3,-2]
abandonments	-17	45826	[-2,-1,-2,-2,-1,-2,-1,-2,-2,-2]
abandons	-13	9	[-2,-1,-1,-2,-1,-2,-1,-2,-1,-2]
abducted	-23	118743	[-3,-1,0,-3,-1,-3,-4,-2,-3,-3]
abduction	-28	87178	[-4,-3,-3,-4,-1,-3,-2,-2,-3,-3]
abductions	-20	141421	[-3,-4,-1,-3,-1,-3,-1,-2,-1,-3]
abhor	-20	109545	[-3,-3,-1,-1,-2,-1,-3,-3,0,-3]
abhorred	-24	149666	[-4,-4,0,-3,-2,-1,-4,-3,-3,0]

Fig. 2. Excerpt from the VADER sentiment lexicon

E. Statistical methods for evaluating reliability of recommendations

We utilized two error quantification measures to determine how closely our recommended results were aligned with the human annotated ground truth:

- Mean Absolute Error (MAE), which measures the average magnitude of errors in the sentiment predictions. Formally,

$$MAE = \frac{1}{N} \sum_i |e_i - v_i| \quad (1)$$

where e_i is the truth i (the human-annotated score in this experiment), while v_i is the estimated value for the same item, and N is the number of items.

- Root Mean Square Error (RMSE): RMSE quantifies deviations from the ground truth, penalizing larger errors more severely. Formally,

$$RMSE = \sqrt{\frac{1}{N} \sum_i (e_i - v_i)^2} \quad (2)$$

These measures (MAE and RMSE) are used to assess whether the auto-generated content is reliable enough to be passed on to a personalization API.

IV. EXPERIMENTAL RESULTS & DISCUSSION

The goal of this section is to provide both qualitative and quantitative assessments of the proposed pipeline. We will begin by providing an overview of the performance of each sentiment engine globally using various error metrics. Next, we will analyze these engines at the aspect level in terms of how they distribute their scores, and then examine the reliability of each.

A. General Overview of System-wide Performance

To provide an adequate basis for evaluating the data, the initial collection of the video material consisted of twenty-three high-quality tech review videos with explicit reference to smartphone products. These were specifically collected from Greek YouTube content. This methodical approach to selecting and collecting these specific types of video content was based on quality parameters to limit speaker overlap within each video segment and other forms of extraneous environmental noises (e.g. background or promotional music)

that may potentially degrade or skew audio captures. Additionally, due to the limited availability of resources related to Greek language multimedia NLP, particularly when it comes to aspect-based sentiment analysis, we believe the size of the current dataset provides sufficient scope for a robust pilot study. However, scaling this corpus to encompass a broader variety of channels and product domains remains a primary focus of our ongoing work.

Table I presents descriptive statistics for the content length of the analyzed videos.

TABLE I. DESCRIPTIVE STATISTICS FOR VIDEO CONTENT

Average number of words	1276.82
Min number of words	384
Max number of words	2137
Standard deviation	509.48

In order to present a general idea of the effectiveness of our system, we started by calculating the average value of several overall performance metrics that were common to both sentiment tracks. As shown in Table II the MAE was approximately 0.278 for the VADER sentiment track after it had been augmented via machine learning-based translation. This indicates a very good agreement in terms of assessment quality when compared to those of human annotators.

The data displayed in Table I were used to determine the reliability of the automated recommendation data for both tracks. Although both tracks performed well within the acceptable range for academic purposes ($MAE < 0.30$), Track 1 was found to have a much higher correlation ($r = 0.84$), with the ground truth established by the manual annotators, than Track 2.

TABLE II. AVERAGED OVERALL PERFORMANCE COMPARISON OF SENTIMENT TRACKS

Sentiment Track	MAE (Mean)	RMSE (Mean)	Correlation (r)
Track 1: Translation + VADER	0.278	0.45	0.84
Track 2: Native Greek (GrAFS)	0.285	0.42	0.71

B. Sentiment Distribution and Emotional Mapping

Regarding the distribution of aspect mentions across the 12 product attributes, the dataset naturally reflects real-world multimedia review patterns. The statistics regarding the presence of aspects in the videos are depicted in Table III. Attributes related to core hardware features, such as ‘Camera’, ‘Battery’, ‘Screen’, ‘Video’, ‘Battery’, ‘Storage space’, ‘Charging’ and ‘Price’, accumulated the highest density of explicit sentiment expressions within the extracted text. Conversely, more specialized features like ‘Sensors’, ‘Fingerprint’ or ‘Processor’ exhibited lower frequencies, as they are typically discussed less frequently or more briefly in standard smartphone video vlogs. This natural imbalance in attribute distribution provides a realistic benchmark for evaluating the adaptability of both lexicon tracks under varying data densities.

The first level of granularity focused on how sentiment was represented across product features. As depicted in Fig. 3, the VADER engine shows a very large number of “Neutral”

and “Compound” sentiments. High “Compound” ratings for the “Camera” and “Screen” categories indicate that there are many subjective opinions about these features that influence the overall sentiment of this feature category in the Greek tech vlog ecosystem.

TABLE III. PRESENCE OF ASPECTS IN VIDEO CONTENT

camera	23
screen	23
price	15
fingerprint	2
battery	20
video	19
design	13
processor	6
charging	16
sensor	13
storage space	15
sound	6

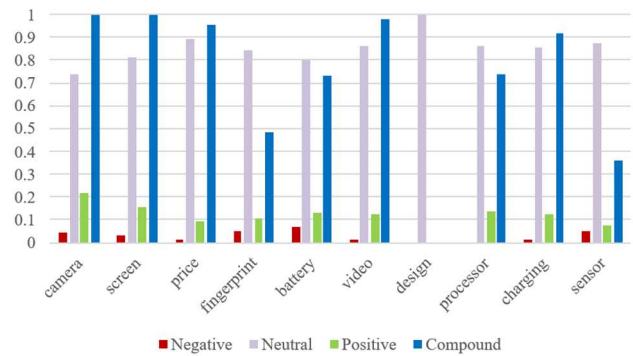


Fig. 3. Sentiment distribution per product aspect using the VADER engine.

Fig. 4 provides a more detailed breakdown of each emotion, based on the native lexicon, which shows that “Happiness” and “Surprise” sentiments are closely related to the “Design” feature category. It is important for RecSys to provide such an emotional understanding when matching a product’s “vibe” with specific user expectations.

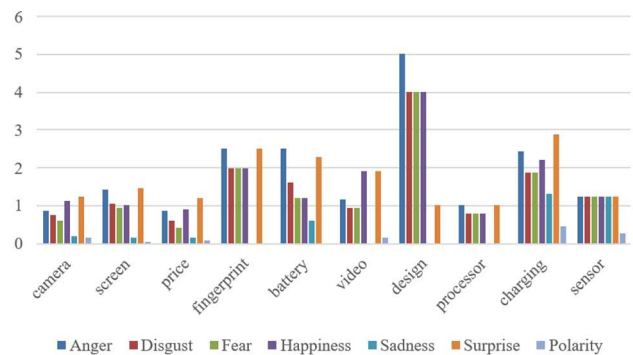


Fig. 4. Granular emotional analysis using the native Greek sentiment lexicon

C. Comparative Accuracy: MAE Evaluation

We use the MAE metric to evaluate the system’s accuracy and the deviation from manually annotated scores. The “average” (AVG) column in Fig. 5 illustrates the same data contained in Table II.

However, since VADER performs consistently, while at the same time some specific aspects like "Screen" display larger deviations from human-annotated scores relative to VADER in the Greek track; this may suggest that translation models are better at processing technical Greek terminology than they are at processing English.

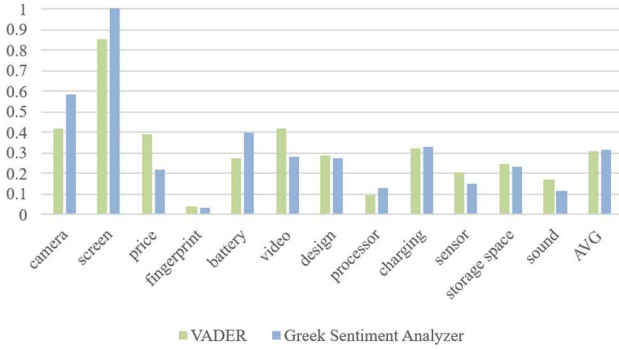


Fig. 5. Comparative MAE between VADER and Greek Sentiment Analyzer.

In order to address data sparseness, we evaluated MAE for only those "non-zero" elements in Fig. 6. When vloggers express their views on an explicit characteristic, the deviation from human-annotated scores increases, especially for "Price" and "Audio". This highlights the difficulties inherent in extracting technical reviews.

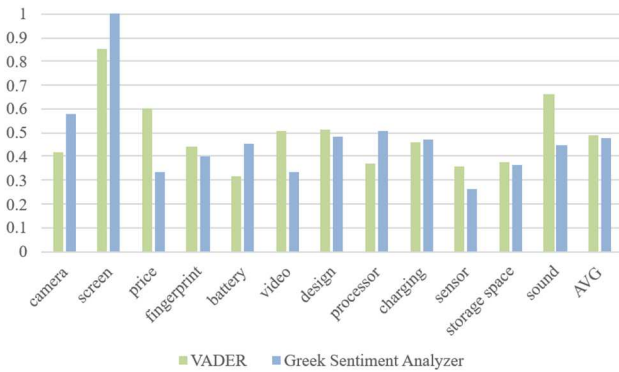


Fig. 6. MAE specifically for non-zero (active) sentiment elements.

D. Reliability and Outlier Analysis (RMSE)

In order to assess the risks of making "catastrophically" wrong predictions, we also employed the RMSE metric, which severely penalizes larger outliers.

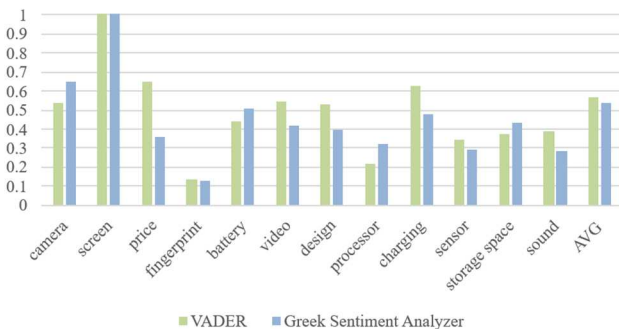


Fig. 7. RMSE per aspect, illustrating system stability.

As depicted in Fig. 7, the average RMSE (AVG column) approximates 0.45. The aspects of "Screen" and "Price" demonstrate the largest RMSE values. These fluctuations imply that recommendations concerning these particular product attributes will require additional semantic filtering to retain user trust.

E. Discussion

The aggregated data from Table II and Fig. 3 through Fig. 7 confirm that the proposed pipeline constitutes a valid source for personalized recommendations. In spite of the noise generated by the ASR layer, the high correlation demonstrated by the VADER track verifies that video/audio content can be converted into a high-fidelity Aspect-Sentiment Matrix for Greece.

To assess the effect of ASR noise in the processing pipeline, reviewers were asked to grade the quality of ASR transcripts. Before this task commenced, a third expert was asked to curate the transcripts of three videos to more accurately reflect the contents of the original sound streams; this step aimed to provide insight on the effect that ASR noise had on the final sentiment analysis outcome. The results of this assessment process are illustrated in Table IV.

TABLE IV. QUALITY OF ASR

	<i>All videos</i>	<i>Non-curated</i>	<i>Curated</i>
Average	0.52	0.4625	0.9
Min	0.225	0.225	0.85
Max	0.95	0.775	0.95
Stddev	0.2	0.147796313	0.04

The data in the Table IV indicate that the quality of ASR is generally deemed mediocre, being in some cases low. This has a definite effect on the quality of aspect-based sentiment analysis, which is supported by the fact that the error magnitudes for the three curated videos were significantly lower than the respective magnitudes in the non-curated ones (a decrease of approximately 30%). Therefore, the improvement of ASR, in conjunction with techniques to appropriately handle technical jargon (or domain-specific issues, in general) can significantly boost aspect-based sentiment assessment accuracy.

While formal statistical significance testing and confidence intervals were not explicitly computed due to the localized size of the review dataset, the high correlation metrics and consistent MAE distributions across the distinct tracks establish a mathematically cohesive evaluation. Future evaluations on expanded datasets will naturally incorporate these validation layers to further secure the statistical robustness of the recommender's core inference steps.

To ground its utility, the generated Aspect-Sentiment Matrix can directly serve as an item repository in content-based recommendation workflows, mapping user preference vectors via metrics like Cosine Similarity to eliminate sentiment masking.

Furthermore, Figs. 5–7 explicitly demonstrate that despite ASR noise, our translation-augmented VADER track maintains a low MAE across aspects. Compared to computationally heavy deep-learning or multilingual transformer baselines, this dual-track pipeline provides a

highly efficient, lightweight alternative tailored for low-resource environments like Modern Greek. Notably, the improvement of the ASR engine suffices to lead to gains of 30% to the accuracy of aspect-based sentiment quantification.

V. CONCLUSIONS, LIMITATIONS AND FUTURE WORK

In this work, we proposed an end-to-end workflow for converting non-structured Greek audio/visual data extracted from YouTube to a structured and feature aware Sentiment Matrix. By using the Vosk ASR for speech-to-text processing and analyzing both dual sentiment tracks -Translation-augmented VADER and the native Greek lexicon- we also showed how feasible it is to automatically extract user preferences for low resource languages such as Modern Greek. Based upon our results, there appears to be a strong relationship ($r=0.84$) between the scores generated by the automatic system and those obtained from human experts. In addition to demonstrating its ability to identify user sentiment related to hardware specific attributes (i.e. Camera, display, battery), we have also shown that the system has particular strength in extracting sentiment associated with other areas where consumers are likely to make their purchasing decisions based on such information.

While the results show significant promise for automated preference extraction for low-resource languages, several major limitations were observed during experimentation:

- **The ASR Bottleneck:** Although the Vosk engine provided a useful baseline, the transcription of technical slang and brand names often resulted in phonetic errors that negatively affected the determination of the correct polarity of user sentiment.
- **Complexity of Pragmatics:** Both engines exhibited difficulty in interpreting sarcasm and irony, common techniques used by Greek vloggers to express opinions about high cost flagship devices and/or software bugs. This difficulty was evidenced by significantly higher RMSE for the “Price” attribute.
- **Data Sparsity:** We observe that when the vlogger discussed niche technical details, the error margin increased (as shown in the non-zero error margin analysis). As a result, it is important for the engines to be augmented with additional specialized domain dictionaries, in order to achieve acceptable accuracy across all 12 product attributes.
- **Dataset Scale:** The localized scale of the review dataset, consisting of 23 high-density video logs, limits immediate statistical generalizability. While sufficient for validating our structural layers and baseline metrics, expanding the corpus across more diverse channels is required to secure broader validation.

Our future work will focus on three main directions:

1. **Transformer-based ASR:** we intend to replace our existing acoustic model with fine-tuned OpenAI Whisper models to improve the transcription of Greek technical terminology and reduce the initial WER [38], [39].
2. **LLM and Transformer Integration:** To provide a stronger comparison with modern baseline

architectures, future versions of this project will cross-evaluate our lexicon tracks against state-of-the-art Transformer models and fine-tuned Large Language Models (e.g., GPT-4 or LLaMA-3) to handle complex pragmatics and irony, thereby benchmarking computational efficiency against increased semantic accuracy [40], [41].

3. **Mobile Recommendation API:** Lastly, we plan to encapsulate this backend in a cross platform mobile application that serves as an AI shopping companion. As a result, consumers will be able to scan product barcode in real time and receive a sentiment summary based on Greek YouTube experts.

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